# Supplementary Archive Figures for the QKD Routing Manuscript

October 2026

This file preserves the project's conceptual and simulator-illustration graphics outside the main manuscript's results flow. The revised main paper uses only figures directly cited in the methods and matched GNN-PPO versus LSTM-PPO analysis. These archive figures provide background context; they are not additional experimental outcomes.

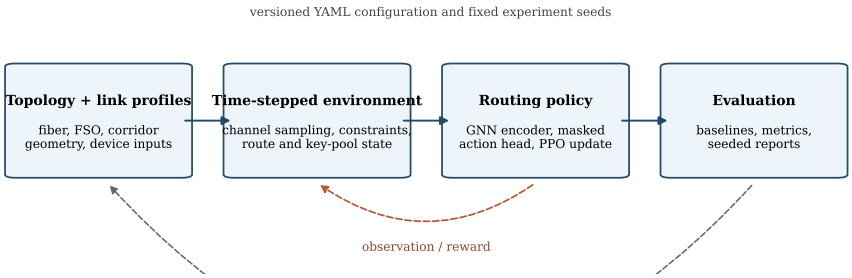



Figure 1: High-level project workflow showing topology and link profiles, time-stepped environment, routing policy, and evaluation. This is an overview diagram rather than a specification of individual model layers.

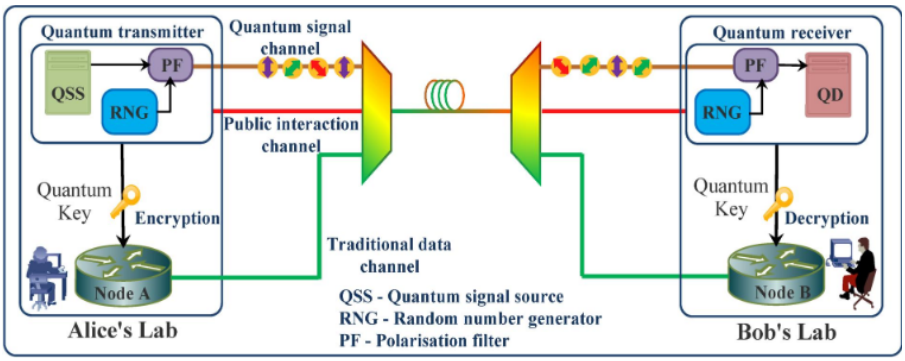



Figure 2: Conceptual distinction between a quantum signal channel and a public classical interaction channel in a prepare-and-measure QKD link. The drawing is educational context, not a hardware diagram of the simulated network.

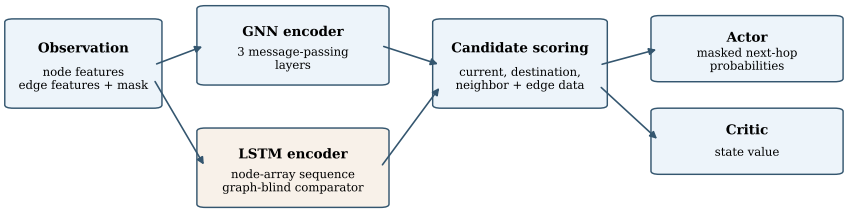



Figure 3: Earlier conceptual GNN-PPO architecture sketch. The implementation-backed actor–critic and feature flow are specified in the main manuscript.

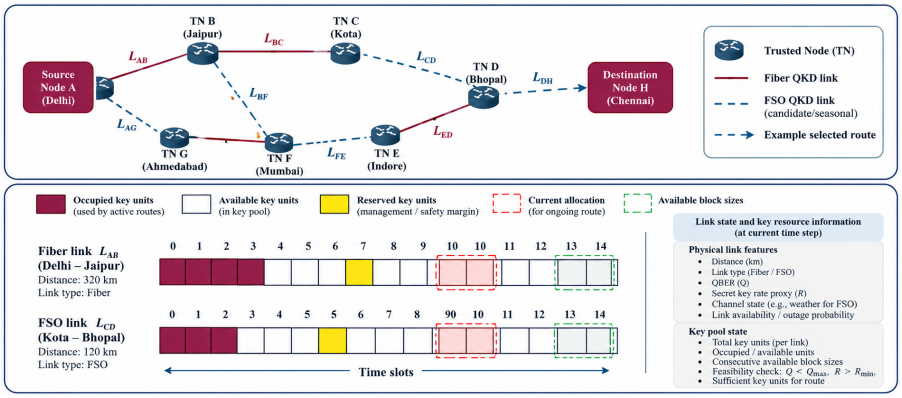



Figure 4: Conceptual hybrid QKD topology and normalized key-pool visualization. Pool states in the simulator are abstractions, not measured link capacities or a calibrated network-wide key-management process.

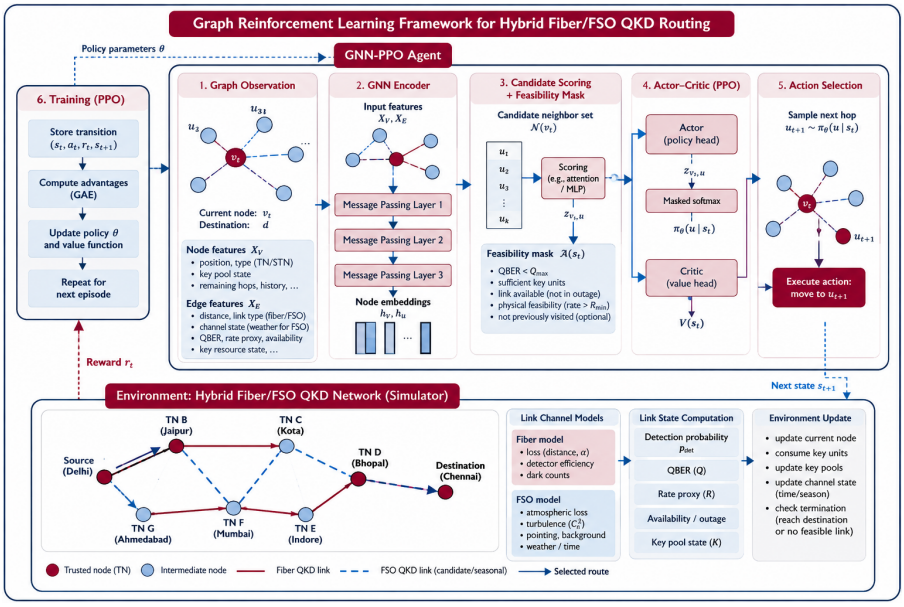



Figure 5: High-level hybrid-routing and reinforcement-learning workflow from an earlier project design. It is a conceptual overview; the final implementation is described in the main manuscript.

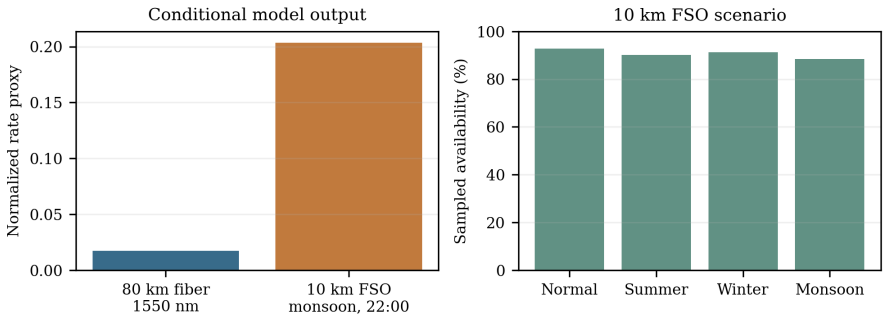



Figure 6: Simulator-generated normalized rate proxy for representative fiber and FSO links and sampled FSO availability across configured seasons. These conditional model outputs are not field or device measurements.

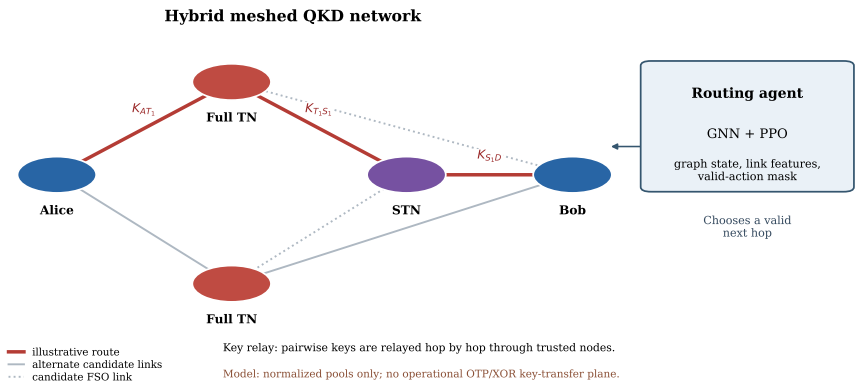



Figure 7: Conceptual meshed trusted-node QKD architecture with direct and relayed paths. The figure illustrates the routing abstraction and does not assert deployed topology or key-throughput measurements.

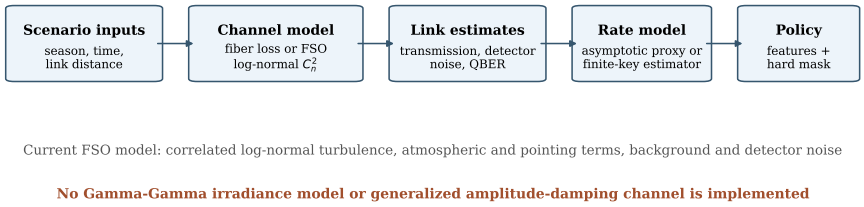



Figure 8: Conceptual mapping from scenario inputs through channel and rate models to a routing decision. Model quantities and evidence limitations are detailed in the main manuscript.

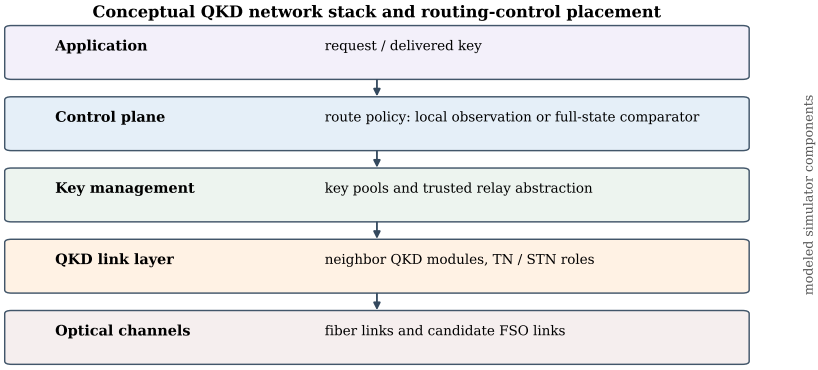



Figure 9: Conceptual QKD network stack placing routing control, key management, trusted-node relaying, and optical channels in context. It is a functional abstraction rather than a deployed protocol stack.

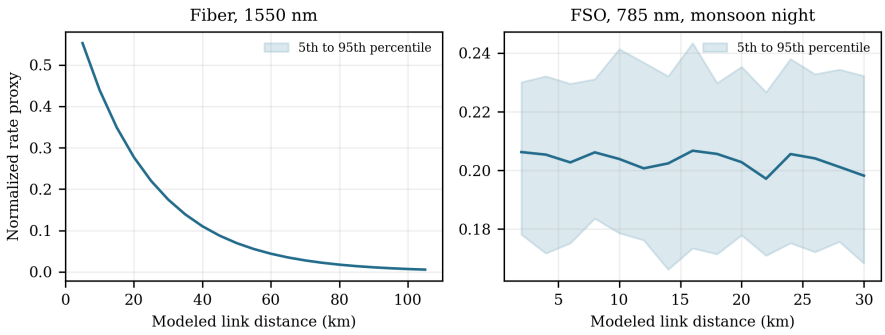



Figure 10: Configured distance-dependent fiber rate proxy and conditional FSO rate proxy with sampled interval bands. These curves summarize simulator assumptions and do not represent calibrated link measurements or secret-key throughput.

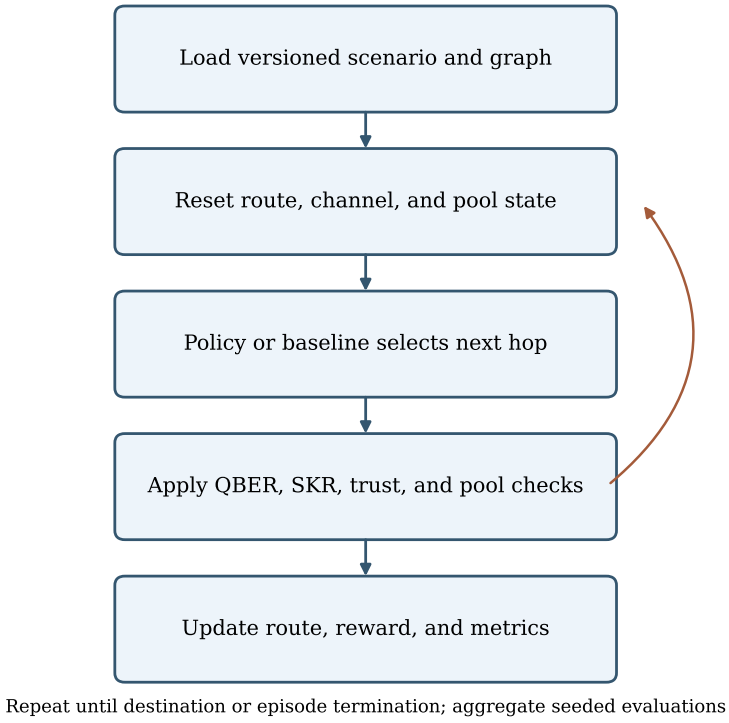



Figure 11: Episode simulation loop: load the graph and scenario, reset route/channel/pool state, select a next hop, apply feasibility and reward updates, and repeat until termination.

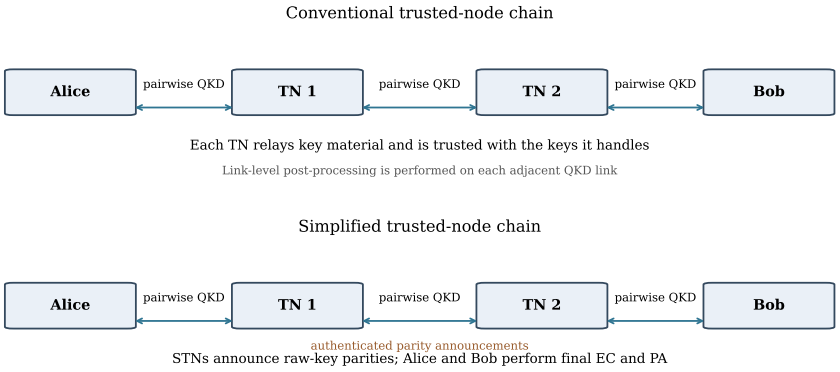



Figure 12: Conceptual comparison of conventional trusted-node and simplified trusted-relay chain abstractions. The diagram is explanatory and does not define an implemented composable security protocol.